CBSE Class 12 Physics — Nuclei

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (MCQ/Assertion-Reason/VSA)

Q1. The curve of binding energy per nucleon as a function of atomic mass number has a sharp peak for the helium nucleus. This implies that the helium nucleus is: (a) radioactive (b) unstable (c) easily fissionable (d) more stable nucleus than its neighbours [PYQ 2023] Very High

Q2. Which of the following statements about nuclear forces is NOT true? (a) The nuclear force between two nucleons falls rapidly to zero as their distance is more than a few femtometres. (b) The nuclear force is much weaker than the Coulomb force. (c) The force is attractive for distances larger than $0.8\ \mathrm{fm}$ and repulsive if they are separated by distances less than $0.8\ \mathrm{fm}$. (d) The nuclear force between neutron-neutron, proton-neutron and proton-proton is approximately the same. [PYQ 2022] Very High

Q3. When two light nuclei ($A \leq 10$) fuse together to form a heavier nucleus, the: (a) binding energy per nucleon increases. (b) binding energy per nucleon decreases. (c) binding energy per nucleon does not change. (d) total binding energy decreases. [PYQ 2020] Very High

Q4. The ratio of the nuclear densities of two nuclei having the mass numbers 8 and 27 is: (a) 8:27 (b) 3:2 (c) 2:3 (d) 1:1 [PYQ 2025-26 SQP] Very High

Q5. Binding energy values of a few nuclei are given below: Lithium-7 ($40.1\ \mathrm{MeV}$), Iron-56 ($492.2\ \mathrm{MeV}$), Lead-206 ($1622\ \mathrm{MeV}$), Deuterium-2 ($2.2\ \mathrm{MeV}$). The least and the most stable nuclei respectively are: (a) Lead and Lithium (b) Lithium and Deuterium (c) Deuterium and Iron (d) Iron and Lead [PYQ 2025-26 SQP] High

Q6. Which of the following statements is INCORRECT for atomic nuclei? (a) As mass number A increases, the density of the nucleus does not change. (b) Heavier nuclei tend to have a larger Z/N ratio as Coulomb forces have a longer range compared to nuclear forces. (c) For nuclei with $A > 100$, the binding energy per nucleon decreases on an average as A increases. (d) Two Lithium nuclei do not combine to form Carbon nuclei at room temperature because Coulomb repulsions do not allow them to come very close. [PYQ 2025-26 SQP] High

Q7. Choose the only correct statement regarding nuclear stability: (a) In a stable nucleus, the number of neutrons $\geq$ number of protons. (b) Nuclei with atomic number greater than 82 show a tendency to fuse. (c) A stable nucleus in general has an even number of protons and an odd number of neutrons. (d) The proton-proton nuclear force $>$ proton-neutron nuclear force $>$ neutron-neutron nuclear force. [PYQ 2025-26 SQP] High

Q8. If M is the actual mass of the nucleus ${}^{27}_{13}Al$, and m_p and m_n are the masses of protons and neutrons respectively, the binding energy of the nucleus is given by: (a) $[13m_p + 14m_n - M]c^2$ (b) $[14m_p + 13m_n - M]c^2$ (c) $[M - 13m_p - 14m_n]c^2$ (d) $[M - 27m_p - 27m_n]c^2$ [PYQ 2025-26 SQP] High

Q9. Complete the following nuclear reaction: ${}^{10}_{5}B + {}^{1}_{0}n \rightarrow {}^{4}_{2}He + ...$ [PYQ 2015] Medium

Q10. Which physical quantity in a nuclear reaction is considered equivalent to the Q-value of the reaction? [PYQ 2020] Medium

Q11. Four nuclei of an element undergo fusion to form a heavier nucleus, with the release of energy. Which of the two — the parent or the daughter nucleus — would have higher binding energy per nucleon? [PYQ 2018] Very High

Q12. Write the empirical relation showing the dependence of the nuclear radius (R) on its mass number (A). State the approximate value of the constant R_0. [Practice] High

Q13. Define the term "mass defect" of a nucleus. [Practice] High

Q14. Express 1 atomic mass unit (amu or u) in terms of its equivalent energy in MeV. [Practice] Very High

Q15. What is the order of magnitude of nuclear density? [Practice] Medium

Q16. Assertion (A): The density of all atomic nuclei is approximately the same. **Reason (R):** The radius of a nucleus is directly proportional to the cube root of its mass number. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) Both A and R are false. [Practice] Very High

Q17. Assertion (A): The nucleus of Iron (${}^{56}Fe$) is one of the most stable nuclei in nature. **Reason (R):** The binding energy per nucleon is maximum for Iron (${}^{56}Fe$), which is approximately $8.8\ \mathrm{MeV}$. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) Both A and R are false. [Practice] Very High

Q18. Assertion (A): The process of nuclear fission of heavy nuclei releases a large amount of energy. **Reason (R):** The binding energy per nucleon of the product nuclei is greater than that of the original heavy nucleus. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) Both A and R are false. [Practice] Very High

Q19. Complete the following nuclear reaction: ${}^{94}_{42}Mo + {}^{2}_{1}H \rightarrow {}^{95}_{43}Tc + \ldots\ldots$ [PYQ 2015] Medium

Q20. Define the term 'isobars' in the context of atomic nuclei. [Practice] Medium

2-Mark Questions

Q21. How is the size of a nucleus found experimentally? Write the mathematical relation between the radius and mass number of a nucleus. [PYQ 2023] Very High

Q22. The nuclear radius of ${}^{27}_{13}Al$ is $3.6\ \mathrm{fm}$. Find the exact nuclear radius of ${}^{64}_{29}Cu$. [PYQ 2020] Very High

Q23. Give two fundamental differences between nuclear fission and nuclear fusion. [PYQ 2022] Very High

Q24. Show mathematically that the density of a nucleus is completely independent of its mass number A. [PYQ 2019] Very High

Q25. State two distinguishing features of the nuclear force. [PYQ 2019] Very High

Q26. Draw a plot showing the variation of potential energy of a pair of nucleons as a function of their separation. Clearly mark the regions on the graph where the force is: (i) attractive, and (ii) repulsive. [PYQ 2019] Very High

Q27. The binding energies of ${}^{16}_{8}O$ and ${}^{35}_{17}Cl$ are $127.35\ \mathrm{MeV}$ and $289.3\ \mathrm{MeV}$ respectively. Which of the two nuclei is more stable and why? [Practice] High

Q28. Give a scientific reason: Why does it take more energy, in general, to remove a neutron from a stable nucleus than to remove a proton? [PYQ 2025-26 SQP] High

Q29. Two nuclei may have the exact same radius, even though they contain different numbers of protons and neutrons. Explain how this is possible, and give the specific name for such a pair of nuclei. [PYQ 2022] High

Q30. Why is the actual mass of a stable nucleus always strictly less than the sum of the masses of its constituent protons and neutrons? What is this difference called? [Practice] High

Q31. What is the 'Q-value' of a nuclear reaction? State the required condition for a nuclear reaction to be spontaneous (exothermic). [Practice] Medium

Q32. Explain why nuclear fusion reactions require extremely high temperatures and pressures to occur. [Practice] High

3-Mark Questions

Q33. A heavy nucleus with mass number $A = 240$ and binding energy per nucleon $\mathrm{BE}/A = 7.6\ \mathrm{MeV}$ breaks into two identical fragments, each of $A = 120$ with $\mathrm{BE}/A = 8.5\ \mathrm{MeV}$. Calculate the exact energy released in this fission process. [PYQ 2016] Very High

Q34. Distinguish between nuclear fission and fusion. Calculate the energy released (in MeV) in the deuterium-tritium fusion reaction: ${}^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + {}^{1}_{0}n$. Given: $m({}^{2}_{1}H) = 2.014102\ \mathrm{u}$, $m({}^{3}_{1}H) = 3.016049\ \mathrm{u}$, $m({}^{4}_{2}He) = 4.002603\ \mathrm{u}$, $m_n = 1.008665\ \mathrm{u}$. $(1\ \mathrm{u} = 931.5\ \mathrm{MeV}/\mathrm{c}^2)$. [PYQ 2015] Very High

Q35. A fast-moving neutron collides with the nucleus of Plutonium (Pu), thereby producing Xenon (Xe) and Zirconium (Zr) along with neutrons. (I) Write the complete balanced nuclear fission reaction. (II) Find the energy released in the above nuclear reaction (Q-value). Given atomic masses: $m({}^{239}_{94}Pu) = 239.052157\ \mathrm{u}$, $m({}^{103}_{40}Zr) = 102.926597\ \mathrm{u}$, $m({}^{134}_{54}Xe) = 133.905040\ \mathrm{u}$, $m({}^{1}_{0}n) = 1.00866\ \mathrm{u}$. [PYQ 2025-26 SQP] Very High

Q36. Explain the processes of nuclear fission and nuclear fusion heavily relying on the plot of binding energy per nucleon (BE/A) versus the mass number A. [PYQ 2018] Very High

Q37. Plot a graph showing the variation of binding energy per nucleon as a function of mass number A. Which specific property of the nuclear force flawlessly explains the approximate constancy of binding energy in the intermediate mass range $30 < A < 170$? [PYQ 2017] Very High

Q38. Calculate the binding energy and the binding energy per nucleon for an Alpha particle (${}^{4}_{2}He$). Given: Mass of proton $m_p = 1.0078\ \mathrm{u}$, Mass of neutron $m_n = 1.0087\ \mathrm{u}$, Mass of Helium nucleus $m_{He} = 4.0026\ \mathrm{u}$. $(1\ \mathrm{u} = 931.5\ \mathrm{MeV})$. [Practice] High

Q39. If both the total number of protons and neutrons in a nuclear reaction are absolutely conserved, in what physical way is mass converted into energy (or vice versa)? Explain giving one example. [PYQ 2015] High

Q40. Calculate the energy equivalent of $1\ \mathrm{gram}$ of pure substance in Joules and in MeV. [Practice] Medium

Q41. Light nuclei readily undergo fusion to form a heavier nucleus, whereas heavy nuclei tend to undergo fission. Give a solid scientific reason based on the binding energy curve. [Practice] Medium

Q42. Differentiate clearly between Isotopes, Isobars, and Isotones. Give one standard example for each. [Practice] High

5-Mark Questions (Long Answer/Case-Study)

Q43. Case Study: Binding Energy and Stability The stability of a nucleus is primarily determined by its binding energy per nucleon (BE/A). The greater the BE/A, the more tightly bound the nucleons are, rendering the nucleus more stable. The BE/A curve peaks near Iron ($A = 56$) and gradually decreases for heavier elements. (A) Identify the most stable nucleus from the following based strictly on the typical BE curve: $^4He, ^{16}O, ^{56}Fe, ^{235}U$. (B) Explain mathematically why a heavy nucleus like Uranium ($A = 235$) tends to undergo fission when struck by a slow neutron. (C) What does the plateau (constancy) in the BE/A curve for $30 < A < 170$ suggest about the nature of the nuclear force? (D) If the average binding energy per nucleon of a nucleus with mass number 60 is $8.5\ \mathrm{MeV}$, what is the total binding energy required to completely dismantle this nucleus into its individual protons and neutrons? [Practice] Very High

Q44. Case Study: Nuclear Fission Nuclear fission is a process in which a heavy nucleus splits into two intermediate-mass fragments of comparable size, accompanied by the release of neutrons and a vast amount of energy. This process is the foundational principle behind nuclear power reactors and atomic weapons. (A) Write the standard balanced nuclear equation for the fission of $^{235}_{92}U$ when bombarded by a slow-moving thermal neutron, producing $^{144}_{56}Ba$ and $^{89}_{36}Kr$. (B) In the above fission process, what is the exact number of secondary neutrons produced? (C) Why does the fission of a heavy nucleus invariably release energy? Explain using the concept of mass defect. (D) What is a nuclear chain reaction, and what essential role do the secondary neutrons play in it? [Practice] Very High

Q45. Case Study: Nuclear Force The strong nuclear force is the fundamental interaction responsible for binding protons and neutrons together within the tiny nucleus, overcoming the immense electrostatic repulsion between the positively charged protons. (A) State any three unique characteristics of the nuclear force that distinguishes it from the Coulomb (electrostatic) force. (B) Draw a graph showing the variation of potential energy of a pair of nucleons as a function of their separation. (C) Based on the graph, at what approximate separation distance is the potential energy minimum, and what does this indicate about the nuclear force at that exact distance? (D) Explain the term "saturation of nuclear forces". [Practice] Very High

Q46. (A) Plot a neat, labelled graph showing the variation of binding energy per nucleon (BE/A) versus mass number A for a large number of nuclei. (B) Use this graph to explicitly explain how energy is released in both the processes of (i) nuclear fission of heavy nuclei, and (ii) nuclear fusion of light nuclei. (C) Why is the nuclear density approximately constant across almost all elements in the periodic table? Give a mathematical proof. [PYQ 2017 / 2019] Very High

Q47. (A) Define nuclear mass defect (Δm) and nuclear binding energy (BE). Write the standard relation between them. (B) A certain nucleus X has a mass number of 125, while another nucleus Y has a mass number of 64. Calculate the exact ratio of their nuclear radii ($R_X : R_Y$). (C) What will be the ratio of their nuclear densities ($\rho_X : \rho_Y$)? Give a reason for your answer. (D) Calculate the exact energy released in MeV if $1\ \textbf{atomic mass unit}\ (\mathbf{u})$ is completely annihilated into energy. [PYQ 2020 / Practice] Very High

Q48. Case Study: Mass-Energy Equivalence & Discovery of Neutron In 1932, James Chadwick performed a landmark experiment by bombarding a Beryllium target with alpha particles, leading to the discovery of the neutron. The emitted neutral radiation was highly penetrating and could eject protons from paraffin wax. (A) Complete the nuclear reaction for Chadwick's experiment: $^9_4Be +^4_2 He \rightarrow^{12}_6 C + ...$ (B) Why did Chadwick conclude that the emitted radiation consisted of neutral particles (neutrons) rather than highly energetic gamma-ray photons? (C) Why are neutrons considered ideal projectile particles for inducing nuclear reactions compared to protons or alpha particles? (D) Calculate the Q-value for a reaction where the total initial rest mass is $10.012\ \mathrm{u}$ and the total final rest mass is $10.008\ \mathrm{u}$. Is the reaction exothermic or endothermic? [PYQ 2022 / Practice] High

Q49. (A) Define binding energy per nucleon. How is it directly related to the absolute stability of a nucleus? (B) Two nuclei of Deuterium (^2_1H) undergo fusion to form a Helium nucleus (^4_2He). If the binding energy per nucleon of Deuterium is $1.1\ \mathrm{MeV}$ and that of Helium is $7.1\ \mathrm{MeV}$, calculate the total energy released in this specific fusion reaction. (C) Despite the huge energy release, why do we not observe Deuterium nuclei spontaneously fusing at normal room temperatures? [Practice] Very High

Q50. (A) State Einstein's mass-energy equivalence principle and write its famous mathematical equation. (B) Calculate the mass defect (Δm) and the total binding energy of an Oxygen-16 ($^{16}_8O$) nucleus. (Given: Mass of proton = $1.007825\ \mathrm{u}$, Mass of neutron = $1.008665\ \mathrm{u}$, Mass of $^{16}_8O$ nucleus = $15.994915\ \mathrm{u}$). (C) Calculate the binding energy per nucleon for $^{16}_8O$. (D) Why does a nuclear fusion reaction generally yield significantly more energy *per unit mass* of the reacting fuel compared to a nuclear fission reaction? [Practice] High

Answer Key

Q1: (d) more stable nucleus than its neighbours. **Q2:** (b) The nuclear force is much weaker than the Coulomb force. (It is actually ~ 100 times stronger). **Q3:** (a) binding energy per nucleon increases. **Q4:** (d) 1:1. (Nuclear density is independent of mass number A). **Q5:** (c) Deuterium and Iron. (BE/A for Li=5.7, Fe=8.78, Pb=7.87, H-2=1.1. Least is H-2, Most is Fe-56). **Q6:** (b) Heavier nuclei tend to have a larger Z/N ratio. (False, they have a larger N/Z ratio to compensate for Coulomb repulsion). **Q7:** (a) In a stable nucleus, the number of neutrons $\geq$ number of protons. **Q8:** (a) $[13m_p + 14m_n - M]c^2$. (Al has $Z = 13$, $N = 14$). **Q9:** ${}^{7}_{3}Li$. (By conservation of mass number $10 + 1 = 4 + A \implies A = 7$, and atomic number $5 + 0 = 2 + Z \implies Z = 3$). **Q10:** The difference in rest mass energy of initial and final particles (Mass defect $\times c^2$), which manifests as the net kinetic energy of the products. **Q11:** The daughter nucleus. (Fusion moves light elements up the BE/A curve, increasing their stability). **Q12:** $R = R_0 A^{1/3}$. $R_0 \approx 1.2 \times 10^{-15}\ \mathrm{m}$ (or $1.2\ \mathrm{fm}$). **Q13:** The difference between the sum of the masses of individual constituent nucleons and the actual mass of the stable nucleus. **Q14:** $1\ \mathrm{u} = 931.5\ \mathrm{MeV}$. **Q15:** $\sim 10^{17}\ \mathrm{kg/m^3}$ (Extremely high). **Q16:** (a) Both A and R are true and R is the correct explanation of A. **Q17:** (a) Both A and R are true and R is the correct explanation of A. **Q18:** (a) Both A and R are true and R is the correct explanation of A. **Q19:** ${}^{1}_{0}n$ (A neutron). (Mass: $94 + 2 = 95 + A \implies A = 1$. Charge: $42 + 1 = 43 + Z \implies Z = 0$). **Q20:** Nuclei that have the same mass number (A) but different atomic numbers (Z). Ex: ${}^{14}C$ and ${}^{14}N$.

Q21: By high-energy electron scattering experiments. Relation: $R = R_0 A^{1/3}$. **Q22:** $R_{Cu} = R_{Al} \times (64/27)^{1/3} = 3.6 \times (4/3) = 4.8\ \mathrm{fm}$. **Q23:** (1) Fission splits a heavy nucleus; Fusion joins light nuclei. (2) Fission can trigger via slow neutrons at room temp; Fusion requires extreme temperature and pressure. **Q24:** Volume $V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi R_0^3 A$. Mass $m \approx A m_n$. Density $\rho = m/V = (A m_n)/(\frac{4}{3}\pi R_0^3 A) = 3m_n/4\pi R_0^3$. Since A cancels out, density is constant. **Q25:** (1) It is the strongest force in nature but has a very short range ($\sim 2 - 3\ \mathrm{fm}$). (2) It is charge-independent (acts equally between p-p, n-n, p-n). **Q26:** Draw a curve dropping to a deep negative well at $r_0 \approx 0.8\ \mathrm{fm}$. Mark $r < r_0$ as strongly repulsive (steep positive rise) and $r > r_0$ as attractive (asymptotically approaching zero). **Q27:** BE/A of O-16 = $127.35/16 = 7.96\ \mathrm{MeV}$. BE/A of Cl-35 = $289.3/35 = 8.26\ \mathrm{MeV}$. Cl-35 is more stable because it has a higher binding energy per nucleon. **Q28:** Protons inside a nucleus experience mutual electrostatic repulsion (Coulomb force) which counteracts the attractive nuclear force, making them slightly easier to remove. Neutrons experience only the attractive nuclear force. **Q29:** Since $R \propto A^{1/3}$, if they have the same mass number A (total protons + neutrons), they have the same radius. Such nuclei are called Isobars (e.g., ${}^{14}_{6}C$ and ${}^{14}_{7}N$). **Q30:** The missing mass (mass defect) is converted into the Binding Energy (Δmc^2) which holds the nucleons tightly together in the nucleus. **Q31:** Q-value is the energy absorbed or released in a reaction. For a spontaneous (exothermic) reaction, $Q > 0$ (Initial mass $>$ Final mass). **Q32:** Light nuclei are positively charged. To fuse, they must overcome huge electrostatic (Coulomb) repulsion to get close enough ($\sim 1\ \mathrm{fm}$) for the attractive nuclear force to dominate. High temps provide the necessary kinetic energy.

Q33: Initial BE = $240 \times 7.6 = 1824\ \mathrm{MeV}$. Final BE = $2 \times (120 \times 8.5) = 240 \times 8.5 = 2040\ \mathrm{MeV}$. Energy released = Final BE - Initial BE = $2040 - 1824 = 216\ \mathrm{MeV}$. **Q34:** Mass defect $\Delta m = (2.014102 + 3.016049) - (4.002603 + 1.008665) = 5.030151 - 5.011268 = 0.018883\ \mathrm{u}$. Energy released $Q = 0.018883 \times 931.5 \approx 17.59\ \mathrm{MeV}$. **Q35:** (I) ${}^{239}_{94}Pu + {}^{1}_{0}n \rightarrow {}^{134}_{54}Xe + {}^{103}_{40}Zr + 3{}^{1}_{0}n$. (II) $\Delta m = (239.052157 + 1.00866) - (133.905040 + 102.926597 + 3 \times 1.00866) = 240.060817 - 239.857617 = 0.2032\ \mathrm{u}$. Q = $0.2032 \times 931.5 = 189.28\ \mathrm{MeV}$. **Q36:** Elements in the middle of the BE/A curve ($A \approx 56$) have the highest BE/A and are most stable. Heavy elements (low BE/A) split into middle elements, increasing their BE/A (Fission). Light elements fuse to form middle elements, increasing their BE/A (Fusion). Both processes shift systems toward higher stability, releasing energy. **Q37:** Draw the standard BE/A curve. The constancy is due to the "saturation" property of the nuclear force—a nucleon only interacts with its nearest neighbours, not all nucleons in the nucleus. **Q38:** $\Delta m = [2(1.0078) + 2(1.0087)] - 4.0026 = [2.0156 + 2.0174] - 4.0026 = 4.0330 - 4.0026 = 0.0304\ \mathrm{u}$. BE = $0.0304 \times 931.5 = 28.31\ \mathrm{MeV}$. BE/A = $28.31/4 = 7.08\ \mathrm{MeV}$. **Q39:** Even if nucleon numbers are conserved, the specific *arrangement* of nucleons changes. Different arrangements have different total binding energies. The difference in rest mass (mass defect) between reactants and products is converted into kinetic energy via $E = \Delta mc^2$. **Q40:** $E = mc^2 = (10^{-3}\ \mathrm{kg}) \times (3 \times 10^8)^2 = 9 \times 10^{13}\ \mathrm{J}$. In MeV: $(9 \times 10^{13})/(1.6 \times 10^{-13}) \approx 5.625 \times 10^{26}\ \mathrm{MeV}$. **Q41:** Light nuclei have low BE/A; fusing them creates a heavier nucleus with a higher BE/A, making it more stable and releasing energy. Heavy nuclei already have too many protons, making them unstable due to huge Coulomb repulsion; adding more particles via fusion is highly unfavourable, so they fission instead. **Q42:** Isotopes: Same Z, different A (${}^{12}C, {}^{14}C$). Isobars: Same A, different Z (${}^{14}C, {}^{14}N$). Isotones: Same number of neutrons ($N = A - Z$), different Z (${}^{14}_{6}C$ has $N = 8$, ${}^{16}_{8}O$ has $N = 8$).

Q43: (A) ${}^{56}Fe$. (B) The BE/A of U-235 is $\sim 7.6\ \mathrm{MeV}$, while fragments have BE/A $\sim 8.5\ \mathrm{MeV}$. Splitting it increases the BE/A by $\sim 0.9\ \mathrm{MeV}$ per nucleon, releasing vast energy. (C) It indicates saturation; the nuclear force is highly short-ranged, so nucleons only bind with immediate neighbours, making total BE strictly proportional to A. (D) Total BE = $A \times (\mathrm{BE}/A) = 60 \times 8.5 = 510\ \mathrm{MeV}$. **Q44:** (A) ${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{144}_{56}Ba + {}^{89}_{36}Kr + 3{}^{1}_{0}n$. (B) Three. (C) The sum of the rest masses of the fragments is less than the mass of the parent nucleus. This mass defect Δm is released as kinetic energy according to $E = \Delta mc^2$. (D) A chain reaction occurs when the secondary neutrons trigger fission in further Uranium nuclei, causing a self-sustaining, exponentially growing release of energy. **Q45:** (A) 1. Much stronger than Coulomb force. 2. Short-range ($\sim 2\ \mathrm{fm}$). 3. Charge independent. (B) Draw a graph dropping steeply to a minimum at $0.8\ \mathrm{fm}$ and rising asymptotically to zero. (C) Minimum at $r_0 \approx 0.8 \text{ fm}$. At this distance, the force transitions from attractive ($r > 0.8 \text{ fm}$) to strongly repulsive ($r < 0.8 \text{ fm}$). (D) Saturation means a nucleon only feels the force from a few closest neighbours, preventing the nucleus from collapsing completely. **Q46:** (A) Plot standard BE/A graph peaking at Fe-56. (B) (i) Fission: Heavy nucleus at far right (low BE/A) moves left toward the peak (higher BE/A), releasing energy. (ii) Fusion: Light nuclei at far left (low BE/A) move right toward the peak (higher BE/A), releasing energy. (C) $\rho = m/V = (Am_n)/(\frac{4}{3}\pi R_0^3 A)$. A explicitly cancels out, yielding a constant density independent of mass number. **Q47:** (A) $\Delta m = [Zm_p + (A-Z)m_n] - M$. $BE = \Delta mc^2$. (B) $R_X/R_Y = (A_X/A_Y)^{1/3} = (125/64)^{1/3} = 5/4 = 1.25$. (C) $1:1$. Nuclear density is essentially constant for all nuclei. (D) $1 \text{ u} = 931.5 \text{ MeV}$. **Q48:** (A) ${}_0^1n$ (A neutron). (B) Gamma rays could not transfer enough momentum to eject heavy protons from paraffin wax. Only a neutral particle with mass comparable to a proton could explain the collision kinematics perfectly. (C) Because they possess zero electric charge, they experience absolutely no Coulomb repulsion when approaching a target nucleus. (D) $\Delta m = 10.012 - 10.008 = 0.004 \text{ u}$. $Q = 0.004 \times 931.5 = 3.726 \text{ MeV}$. Since mass decreases ($Q > 0$), it is highly exothermic. **Q49:** (A) BE/A is the average energy required to remove a single nucleon. Higher BE/A tightly binds nucleons, ensuring higher stability. (B) Initial BE = $2 \times (2 \times 1.1) = 4.4 \text{ MeV}$. Final BE = $4 \times 7.1 = 28.4 \text{ MeV}$. Energy released = $28.4 - 4.4 = 24.0 \text{ MeV}$. (C) Electrostatic repulsion between the positive protons creates a massive Coulomb barrier that prevents them from approaching within the strong force range at room temp. **Q50:** (A) Mass and energy are interconvertible. $E = mc^2$. (B) $\Delta m = [8(1.007825) + 8(1.008665)] - 15.994915 = [8.0626 + 8.06932] - 15.994915 = 16.13192 - 15.994915 = 0.13700$. BE = $0.137005 \times 931.5 = 127.62 \text{ MeV}$. (C) BE/A = $127.62/16 = 7.97 \text{ MeV}$. (D) Light nuclei have a much larger fractional change in BE/A during fusion than heavy nuclei do during fission. Because they are so light, you fit exponentially more fusion events into 1 kg of fuel than fission events into 1 kg of Uranium, resulting in much higher energy density.

CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).